

Review Session 2

Probability, Bayes, Expected Value & Information

Read the question first. Extract the pieces yourself.

Useful tools

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A \mid B)P(B)$$

$$E[X] = \sum_x xP(X = x)$$

$$EVPI = EV_{\text{perfect information}} - EV_{\text{without information}}$$

$$EVSI = EV_{\text{with sample information}} - EV_{\text{without information}}$$

TYPE 1

PROBABILITY RELATIONSHIPS

Union, intersection, complements, independence, and conditional probability.

Do not start with the numbers.

Name the events, draw the Venn diagram if useful, and decide what region the question is asking about.

Question 2**Union / Intersection**

The probability that a visit to a primary care physician's (PCP) office results in neither lab work nor referral to a specialist is 30%. Of those coming to a PCP's office, 45% are referred to specialists and 35% require lab work.

What is the probability that a visit to a PCP's office results in both lab work and referral to a specialist?

0.10,	0.35,	0.25,	0.05,	0.12
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Setup / Work

Answer: _____

Question 4**Independence**

If $P(A) = 0.40$, $P(B) = 0.50$, and $P(A \cap B) = 0.20$; then which one of the following is true?

- A. events A and B can not occur together
- B. events A and B are dependent events
- C. events A and B are independent events
- D. Not enough information is given to answer this question.

Setup / Work

Answer: _____

Question 20**Three-Set Inclusion–Exclusion**

A survey of a group's viewing habits over the last year revealed the following information:

- (i) 31% watched gymnastics
- (ii) 25% watched baseball
- (iii) 21% watched soccer
- (iv) 13% watched gymnastics and baseball
- (v) 11% watched baseball and soccer
- (vi) 9% watched gymnastics and soccer
- (vii) 6% watched all three sports.

Calculate the percentage of the group that watched none of the three sports during the last year.

36,	50,	60,	52,	48
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Setup / Work

Answer: _____

Question 21**Conditional Probability from a Union**

If $P(A) = 0.70$, $P(B) = 0.50$, and $P(A \cup B) = 0.95$, then $P(B | A) = ?$

0.357,	0.750,	0.50,	Not enough information
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Setup / Work

Answer: _____

Question 26**Intersection from Conditional Probability**

The probability that a married man will vote in a given election is 0.50, the probability that a married woman will vote in the election is 0.60, and the probability that a married woman will vote in the election given that her husband votes is 0.90.

What is the probability that a husband and wife will both vote in the election?

0.54,	0.55,	0.46,	0.45
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Setup / Work

Answer: _____

TYPE 2

BAYES' RULE

The information is given in one direction and the question asks in the other.

Read the story first. Identify the prior, the conditional information, and what the question is conditioning on.

The notation belongs in the work box, not in the question.

Question 8**Bayes' Rule / Multiple Choice**

In answering a question on a multiple-choice test a student either knows the answer or guesses. She knows the answer with the probability $2/3$ and guesses with the probability $1/3$. There are 5 multiple-choice alternatives for each question. In other words, the probability that the guessed answer is correct is $1/5$.

What is the conditional probability that a student knew the answer to a question given that she answered it correctly?

$\frac{5}{11},$	$\frac{3}{10},$	$\frac{10}{11},$	$\frac{3}{11}$
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Setup / Work

Answer: _____

Question 7**Bayes' Rule / False Positive**

Suppose a metal detector alarm lights up with a probability of 0.99 if there is a metal object present. However, even if there is no metal object (false positive), the metal detector alarm still lights up with a probability of 0.05. Suppose the probability of a metal object being present is 0.1.

What is the probability that there is no metal object given that the alarm lights up?

0.9000,	0.3125,	0.6875,	0.0500
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Setup / Work

Answer: _____

Question 6**Bayes' Rule**

Upon arrival at a hospital's emergency room, patients are categorized according to their condition as critical, serious, or stable. In the past year:

- (i) 10% of the emergency room patients were critical;
- (ii) 20% of the emergency room patients were serious;
- (iii) the rest of the emergency room patients were stable;
- (iv) 30% of the critical patients died;
- (v) 15% of the serious patients died; and
- (vi) 2% of the stable patients died.

Given that a patient survived, what is the probability that the patient was categorized as serious upon arrival?

0.29,	0.49,	0.18,	0.39,	0.64
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Setup / Work

Answer: _____

Question 13**Reverse Bayes**

In the anticipation of a rain-free outdoor wedding, Alex and Taylor decide to tie the knot in a desert region known for an average of 10 rainy days per year. Unfortunately, the weather forecast predicts rain for tomorrow, the day of Alex and Taylor's wedding. However, the weather forecaster is not entirely accurate. When it actually rains, the forecaster predicts rain 90% of the time, and when it is dry, the forecaster still (erroneously) predicts rain $X\%$ of the time. Given this information, you calculate the probability that it will rain during Alex and Taylor's wedding as 0.30. Assume that there are 365 days in a year. Find X .

0.3000,	0.5000,	0.0592,	0.1000
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Setup / Work**Answer:** _____

TYPE 3

REPEATED TRIALS & DISTRIBUTIONS

The same random process happens several times.

Look for words like independent, five consecutive games, total, minimum, maximum, at least, or at most.

First decide what event across all repetitions must happen.

Question 3**Independent Failures**

The James Webb Space Telescope (JWST) is a complex space observatory with 344 identified single points of failure in its systems. Each of these points of failure has an independent probability of causing the mission to fail, and these probabilities are constant and do not change over the course of the mission.

Given that the probability of any single point of failure causing the JWST mission to fail is 0.0015 (or 0.15%), what is the probability that the JWST mission will be successful and operational?

0.0015,	0.9966,	0.5967,	0.0052,	0.4033
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Setup / Work

Answer: _____

Question 12**Distribution of a Minimum**

Suppose that an employee working at Bell Center walks around the stands selling hot dogs. Assume that in each game, the employee will make either \$150, \$100, \$50 or \$25 selling hot dogs with probabilities 0.05, .35, .45, and .15 respectively. Let the random variable X represent the profit made by the employee selling hot dogs. Suppose that this employee works for 5 consecutive games and reports you only the minimum profit among the five games.

What is the probability of minimum profit in 5 consecutive games being equal to \$50?

0.444,	0.090,	0.850,	0.556
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Setup / Work

Answer: _____

Question 5**Distribution of a Sum**

The probability distribution for the daily sales at Michael's Co. is given below. Assume that the sales on each weekday (from Monday to Friday) are independent from each other.

Daily Sales (\$1,000s)	40	50	60	70
Probability	0.10	0.40	0.30	0.20

What is the probability of total sales between Monday (inclusive) and Friday (inclusive) being at least \$240,000 and at most \$260,000?

Hint: Use Excel or Python to generate 100 random numbers according to the above distribution and compute the probability. Choose the one closest to your answer.

0.22,	0.55,	1.0,	0.95
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Setup / Work

Answer: _____

TYPE 4

DECISION ANALYSIS

Choose an action under uncertainty.

Read for three things:

actions	states of nature	payoffs / losses
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Then ask whether the question gives no information, perfect information, or an imperfect forecast.

Question 1**Insurance / EMV**

A householder is currently considering insuring the contents of her house against theft for one year. She estimates that the contents of her house would cost her \$20,000 to replace.

Local crime statistics indicate that there is a probability of 0.03 that her house will be broken into in the coming year. In that event her losses would be 10%, 20%, or 40% of the contents with probabilities 0.50, 0.35 and 0.15 respectively.

An insurance policy from company A costs \$150 per year but guarantees to replace any losses due to theft.

An insurance policy from company B is cheaper at \$100 per year but the householder has to pay the first \$50 of any loss herself.

Assume that there can be at most one theft per year, and the householder must purchase an insurance policy.

What is the EMV (Expected Monetary Value) of the best option for the householder?

-150,	-101.5,	-108,	-139.8
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Setup / Work

Answer: _____

Question 15**Expected Monetary Value**

Suppose you have three investment options: Option 1 (high-risk stock), Option 2 (low-risk stock), and Option 3 (savings account). The payoffs for each option depend on the market condition: “Market is Up,” “Market is Flat,” and “Market is Down,” each with specific probabilities.

	Market is Up (0.5)	Market is Flat (0.3)	Market is Down (0.2)
Option 1 (high-risk stock)	1500	100	-1000
Option 2 (low-risk stock)	1000	200	-100
Option 3 (savings account)	500	500	500

What decision would you make to maximize your expected returns?

Option 1,	Option 2,	Option 3
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Setup / Work

Answer: _____

Question 10**EVPI**

Consider the following payoff table

Alternative	S_1	S_2	S_3
A_1	50	90	-80
A_2	10	30	-30
A_3	25	70	-10
Prior Probability	0.6	0.3	0.1

What is EVPI (Expected Value of Perfect Information)?

56,	35,	7,	49,	12
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Setup / Work

Answer: _____

Question 16**Perfect Information**

Suppose you have three investment options: Option 1 (high-risk stock), Option 2 (low-risk stock), and Option 3 (savings account). The payoffs for each option depend on the market condition: “Market is Up,” “Market is Flat,” and “Market is Down,” each with specific probabilities.

	Market is Up (0.5)	Market is Flat (0.3)	Market is Down (0.2)
Option 1 (high-risk stock)	1500	100	-1000
Option 2 (low-risk stock)	1000	200	-100
Option 3 (savings account)	500	500	500

What is the value of information about the market condition before making an investment decision?

580,	500,	480,	460,	420
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Setup / Work

Answer: _____

Question 11

Value of Sample Information

Moreno is the credit manager for the Fabritek, one of the largest textile supplier in the region. He is currently faced with the question of whether to extend \$100,000 credit to a potential new customer, a dress manufacturer. Moreno has three categories for the credit-worthiness of a company: poor risk, average risk, and good risk, but he does not know which category fits this potential customer. Experience indicates that 30 percent of companies similar to this dress manufacturer are poor risks, 60 percent are average risks, and 10 percent are good risks. If credit is extended, the expected profit for poor risks is -\$30,000, for average risks \$15,000, and for good risks \$25,000. If credit is not extended, the dress manufacturer will turn to another supplier. Moreno is able to consult a credit-rating organization for a fee of \$X per company evaluated. For companies whose actual credit record turns out to fall into each of the three categories, the following table shows the percentages that were given each of the three possible credit evaluations by the credit-rating organization.

Credit Evaluation	Actual Credit Record		
	Poor	Average	Good
Poor	50%	40%	20%
Average	40%	50%	40%
Good	10%	10%	40%

What is the maximum amount Moreno should pay as a consulting fee for the credit-rating organization?

$$X = 2500, \quad X = 2900, \quad X = 400, \quad X = 950$$

Setup / Work

Answer: _____

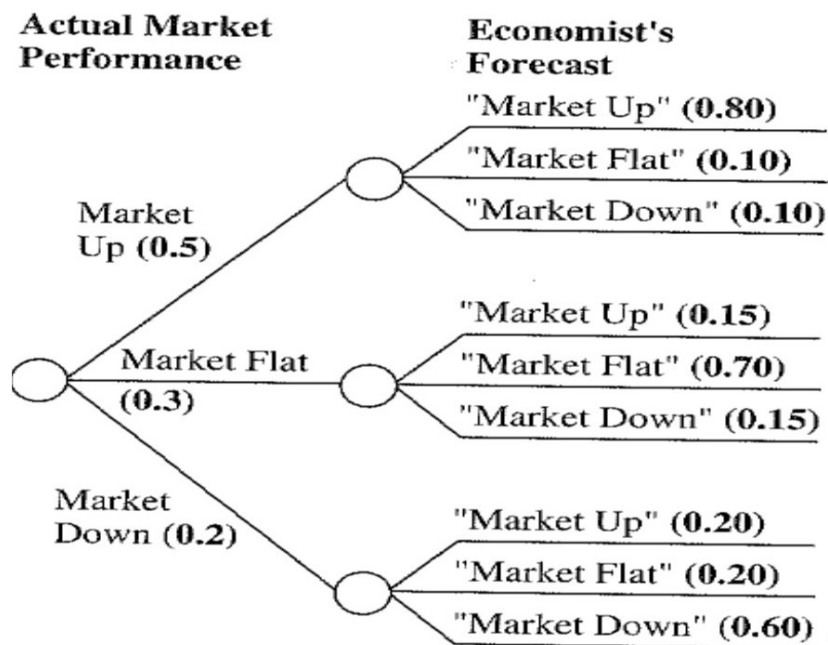
Question 17

Imperfect Information / Economist

Suppose you have three investment options: Option 1 (high-risk stock), Option 2 (low-risk stock), and Option 3 (savings account). The payoffs for each option depend on the market condition: "Market is Up," "Market is Flat," and "Market is Down," each with specific probabilities.

	Market is Up (0.5)	Market is Flat (0.3)	Market is Down (0.2)
Option 1 (high-risk stock)	1500	100	-1000
Option 2 (low-risk stock)	1000	200	-100
Option 3 (savings account)	500	500	500

Suppose that you consult an economist before making your investment decision. Suppose that you have the following probabilities from the past performance record of the economist consultant.



What is the maximum you would be willing to pay for a forecast from this economist before making an investment decision?

822, 540, 242, 580, 350

Setup / Work

Answer: _____

FINAL SECTION

MIXED EXAM REVIEW

The type labels are gone.

For every question, first identify what kind of problem it is.

Do not search for a formula from the numbers. Read, extract, set up, then calculate.

Question 23**Mixed Review**

McGill Marketing decides whether to use direct marketing campaign to reach out the customers.

From the historical data, 45% of the customers request the service by direct marketing contact and the rest request it on their own initiative.

Also, it is estimated that if a customer request the service by direct marketing contact it is more likely that the agent will convince her to purchase premium cable. Exact probabilities conditional on the type of the request is given by

- $P(\text{basic cable} \mid \text{direct marketing}) = 0.05$
- $P(\text{premium} \mid \text{direct marketing}) = 0.95$
- $P(\text{basic cable} \mid \text{own initiative}) = 0.85$
- $P(\text{premium} \mid \text{own initiative}) = 0.15$

What proportion of Premium service customers subscribed via Direct Marketing?

0.427,	0.838,	0.659,	0.270
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Setup / Work

Answer: _____

Question 19**Mixed Review**

A householder is currently considering insuring the contents of her house against theft for one year. She estimates that the contents of her house would cost her \$20,000 to replace.

Local crime statistics indicate that there is a probability of 0.03 that her house will be broken into in the coming year. In that event her losses would be 10%, 20%, or 40% of the contents with probabilities 0.50, 0.35 and 0.15 respectively.

An insurance policy from company A costs \$150 per year but guarantees to replace any losses due to theft.

An insurance policy from company B is even cheaper at \$75 per year but only replaces a fraction (40%) of any loss suffered.

Assume that there can be at most one theft per year, and the householder must purchase an insurance policy.

What is the EMV (Expected Monetary Value) of the best option for the householder?

-150,	-139.8,	-108,	-101.5
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Setup / Work

Answer: _____

Question 18**Mixed Review**

Suppose that we roll a die twice. Consider the following two events:

- E_1 = Second roll is 5.
- E_2 = Difference between the two rolls is 4.

Match each probability with the appropriate answer.

$$P(E_1), \quad P(E_2), \quad P(E_1 \text{ or } E_2), \quad P(E_1 \text{ and } E_2), \quad P(E_1 | E_2), \quad P(E_2 | E_1)$$

$\frac{1}{4},$	$\frac{1}{6},$	$\frac{1}{9},$	$\frac{1}{36}$
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Setup / Work

Answer: _____

Question 27**Mixed Review**

Suppose that an employee working at Bell Center walks around the stands selling hot dogs. Assume that in each game, the employee will make either \$100, \$80, \$50 or \$15 selling hot dogs with probabilities 0.15, .25, .35, and .25 respectively. Let the random variable X represent the profit made by the employee selling hot dogs. Suppose that this employee works for 5 consecutive games and reports you only the maximum profit among the five games.

What is the probability of maximum profit in 5 consecutive games being equal to \$50?

0.922,	0.005,	0.600,	0.078
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Setup / Work

Answer: _____

Question 9**Mixed Review**

The probability that a married man will vote in a given election is 0.50, the probability that a married woman will vote in the election is 0.60, and the probability that a married woman will vote in the election given that her husband votes is 0.90.

What is the probability that at least one of the married couple will vote in the election?

0.54,	0.45,	0.35,	0.65
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Setup / Work

Answer: _____

Question 22**Mixed Review**

The probability distribution for the daily sales at Michael's Co. is given below. Assume that the sales on each weekday (from Monday to Friday) are independent from each other.

Daily Sales (\$1,000s)	40	50	60	70
Probability	0.10	0.40	0.30	0.20

What is the probability of total sales between Monday (inclusive) and Friday (inclusive) being at least \$280,000?

Hint: Use Excel or Python to generate 100 random numbers according to the above distribution and compute the probability. Choose the one closest to your answer.

0.90,	0.10,	0.30,	0.59
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Setup / Work

Answer: _____

Question 24**Mixed Review**

Suppose a metal detector alarm lights up with a probability of 0.99 if there is a metal object present. However, even if there is no metal object (false positive), the metal detector alarm still lights up with a probability of 0.05. Suppose the probability of a metal object being present is 0.1.

What is the probability that there is metal object given that the alarm lights up?

0.1000,	0.3125,	0.6875,	0.9900
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Setup / Work

Answer: _____

Question 14**Mixed Review**

After the contestant randomly selects one door out of three, with the possibility of a car behind it, the game show host, who is aware of the contents behind each door, deliberately opens another door that was not chosen by the contestant, and asks the contestant whether she wants to switch doors. Note that knowing what is behind each door, the host always reveals a goat behind this opened door.

Now calculate the probability that the contestant gets the car if the contestant switches doors.

$\frac{1}{3},$	$\frac{1}{2},$	$\frac{2}{3},$	$0,$	1
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Setup / Work

Answer: _____

Question 25**Mixed Review**

Consider two urns. The first contains three white and six black balls, and the second contains four white and five black balls. We flip a fair coin and then draw a ball from the first urn or the second urn depending on whether the outcome was heads or tails.

What is the conditional probability that the outcome of the toss was heads given that a white ball was selected?

$\frac{3}{9}$,	$\frac{3}{7}$,	$\frac{7}{9}$,	$\frac{22}{67}$,	None of above answers
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Setup / Work

Answer: _____